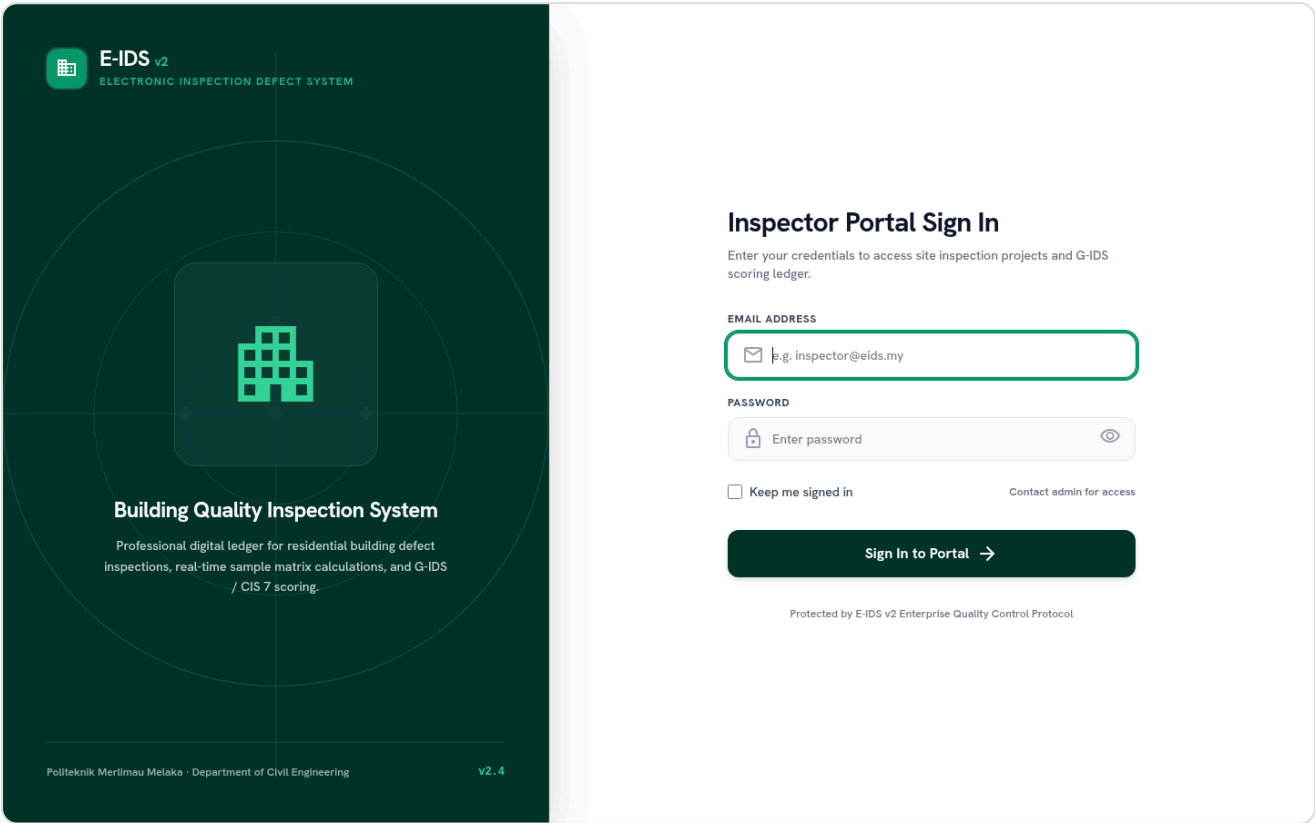


E-IDS v2 — User Manual & Client Guide

Electronic Inspection Defect System (E-IDS v2) *Official Operational Manual for Admins, Site Inspectors, and Contractors*

Updated 2026-08-02 to reflect the simplified 3-role system (Admin, Inspector, Contractor), per-role project/defect visibility, and the 4-state defect lifecycle.

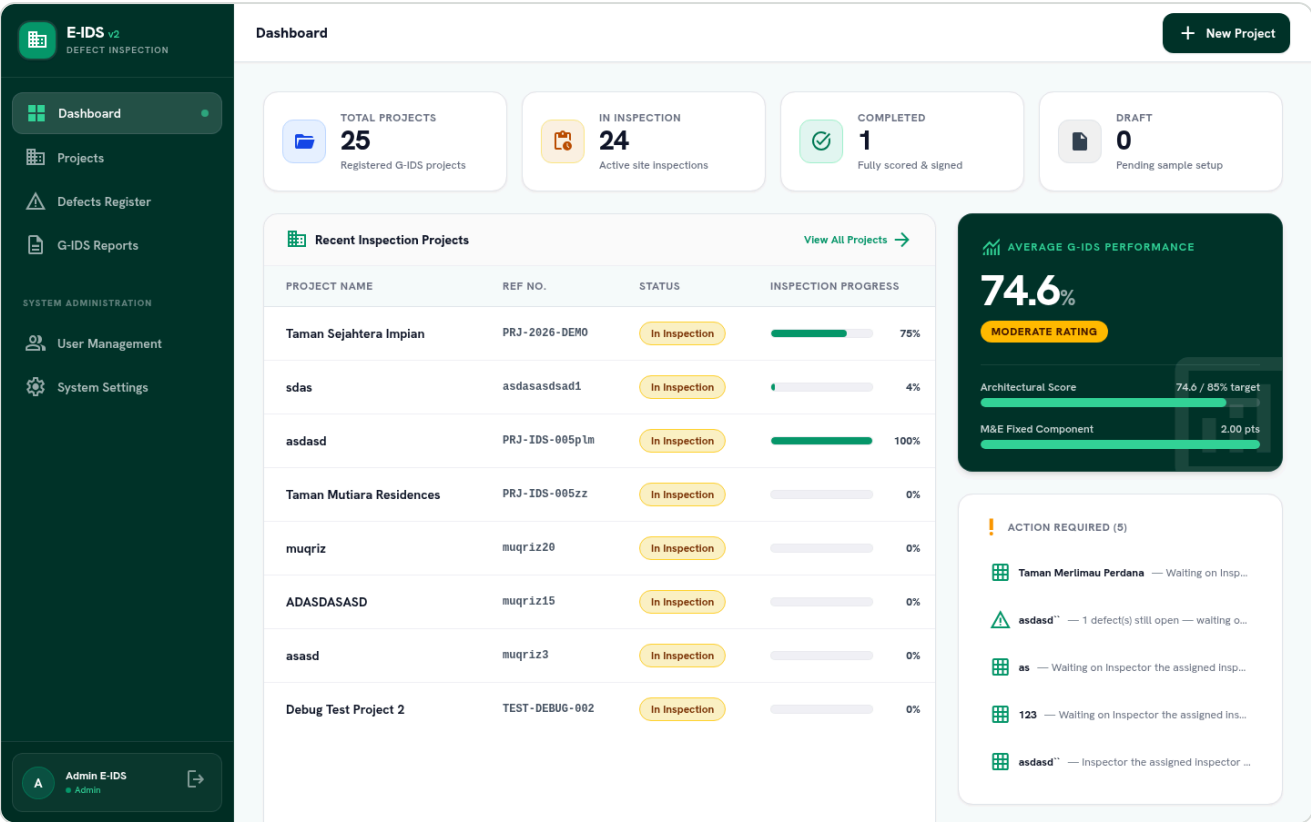


The sign-in screen every role shares — the same login form routes Admin, Inspector, and Contractor to their own scoped view of the system.

1. System Overview

E-IDS v2 is a specialized web-based building defect inspection and scoring platform for the Malaysian construction sector. Built in compliance with G-IDS / CIS 7 quality standards, E-IDS replaces manual paper-based forms with an end-to-end digital workflow:





The Dashboard is the landing page for Admin and Inspector — project totals, the average score across every visible project, and an Action Required list of what still needs attention.

2. User Roles & Permission Matrix

E-IDS v2 supports **three** roles. Each role's view of the system is scoped — Admin sees every project; Inspector and Contractor see only what they're individually assigned to.

Contractor / Developer (Contractor)			Field Inspector (Inspector)			System Admin		
Log Defects Only — No Dashboard/Projects Access	Mark Requests in Progress & Detail	Cancel Self-Cancelled Requests	View Only Their Assigned Projects	Generate Setup & P-Point Component Grid Checks	Verify Contractor Requests & Close Defects	User Management & Roles	Formula Settings & Templates	View & Manage Every Project
						Register Projects, Assign Inspectors/Contractors		Sign Off Completion & Generate the Signed PDF

Detailed Permissions Table

ACTION / CAPABILITY	ADMIN	INSPECTOR	CONTRACTOR
View Dashboard, Projects & Reports	✓ all	✓ assigned	✗
Download Official PDF Report	✓ all	✗ (view score only)	✗
Register & Edit Projects	✓	✓	✗
Assign the Inspector (<code>assigned_to</code>)	✓	✗ (locked to self)	✗
Assign the Contractor (<code>assigned_contractor_id</code>)	✓	✓	✗
Configure Sample Location Names	✓	✓	✗
Perform 5-Point Checks (<code>PASS</code> / <code>FAIL</code>)	✗ (handed off to Inspector)	✓	✗
Upload Photo Evidence (Spatie Media)	✗	✓	✗
View Defects Register / My Defects	✓ all	✓ assigned projects	✓ own-assigned project only
Create / Manually Log a Defect	✓	✓	✗
Mark Defect <code>IN_PROGRESS</code> ("Start Repair")	✓	✓	✓
Mark <code>IN_PROGRESS</code> → <code>PENDING_VERIFICATION</code> ("Mark Settled")	✓	✓	✓
Confirm <code>RESOLVED</code> or Reject back to <code>IN_PROGRESS</code>	✓	✓	✗
Mark Project as Completed (sign-off)	✓	✗	✗
Manage Users & Role Access	✓	✗	✗
Modify G-IDS Formula Settings	✓	✗	✗
Delete Project (with Modal Confirm)	✓	✗	✗

What "scoped" means in practice: opening a project or defect you don't have visibility into — whether from a list or by typing the URL directly — returns a "Forbidden" page for every role except Admin. This applies uniformly across Projects, the Dashboard, Defects, and Reports.

E-IDS v2

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Projects

+ New Project

Search project name, number or developer...

All Statuses

All Building Types

All Inspectors

Filter

PROJECT	REF NO.	TYPE	STATUS	PROGRESS	SCORE	ACTIONS
Taman Sejahtera Impian Implan Sentosa Development Sdn Bhd	PRJ-2026-DEMO	Terrace	In Inspection	75%	84.8%	
sdas asdas	asdasadsad1	Terrace	In Inspection	4%	31.8%	
asdasd Mutlara Development Sdn Bhd	PRJ-IDS-005p1m	Semi-D	In Inspection	100%	102.8%	
Taman Mutiara Residences asdasd	PRJ-IDS-005zz	Bungalow	In Inspection	0%	—	
muqriz asdasd	muqriz20	Terrace	In Inspection	0%	—	
ADASDASAD asdasd	muqriz15	Semi-D	In Inspection	0%	—	
asasd aadasdad	muqriz3	Terrace	In Inspection	0%	—	
Debug Test Project 2 Test Dev	TEST-DEBUG-002	Terrace	In Inspection	0%	—	
as as	muqriz2	Semi-D	In Inspection	0%	—	

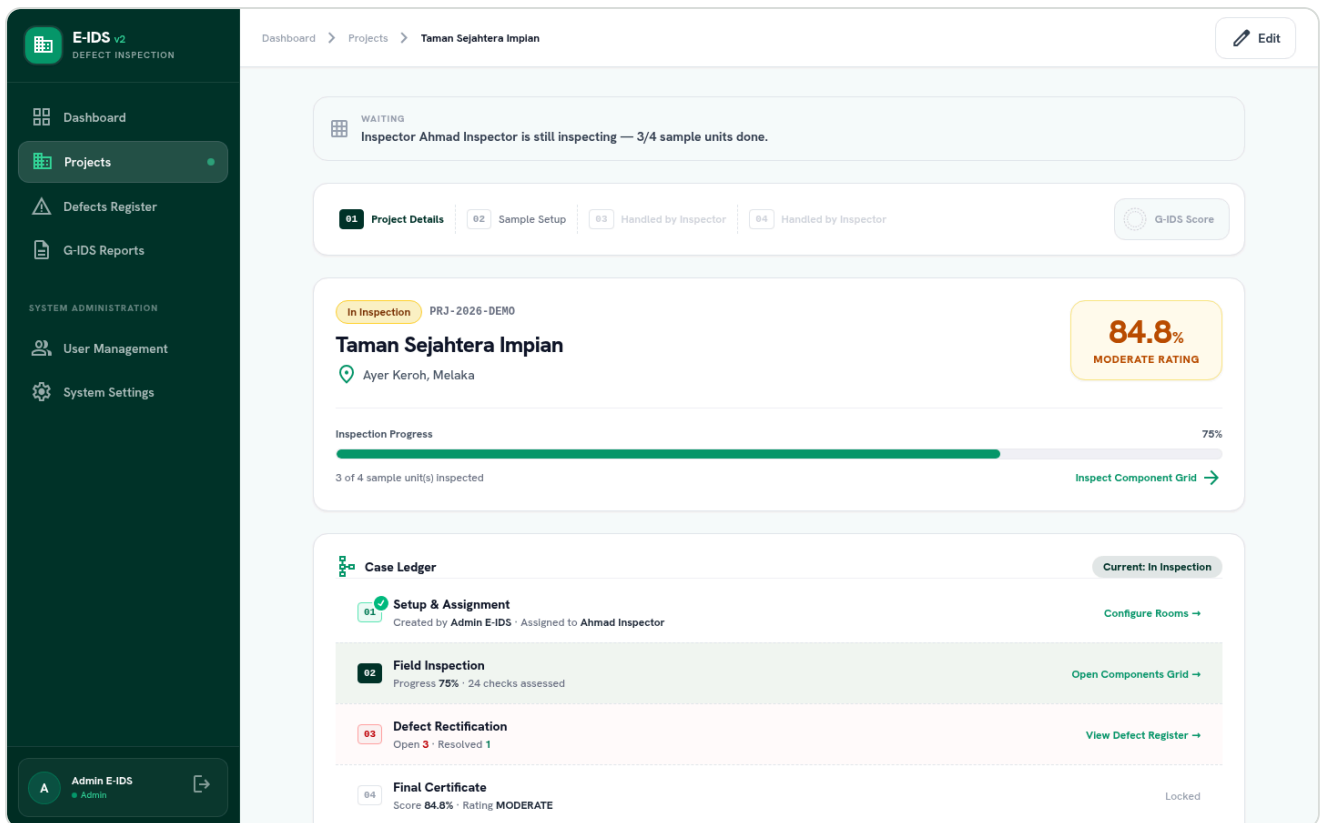
The Projects list — the same page renders different rows for each role: Admin sees every project, Inspector and Contractor only ever see rows they're assigned to.

3. Operational Role Handoff & Project Lifecycle

E-IDS v2 enforces a clear phase handoff between project creation, site inspection, contractor repairs, and final certificate sign-off. Admin's part of the workflow pauses once sample setup is saved — from there, the assigned Inspector and Contractor take over independently, each seeing only what's relevant to them — before Admin returns at the end to sign off and generate the certificate:

```
sequenceDiagram
    participant Admin
    participant Inspector as Assigned Inspector
    participant Contractor as Assigned Contractor

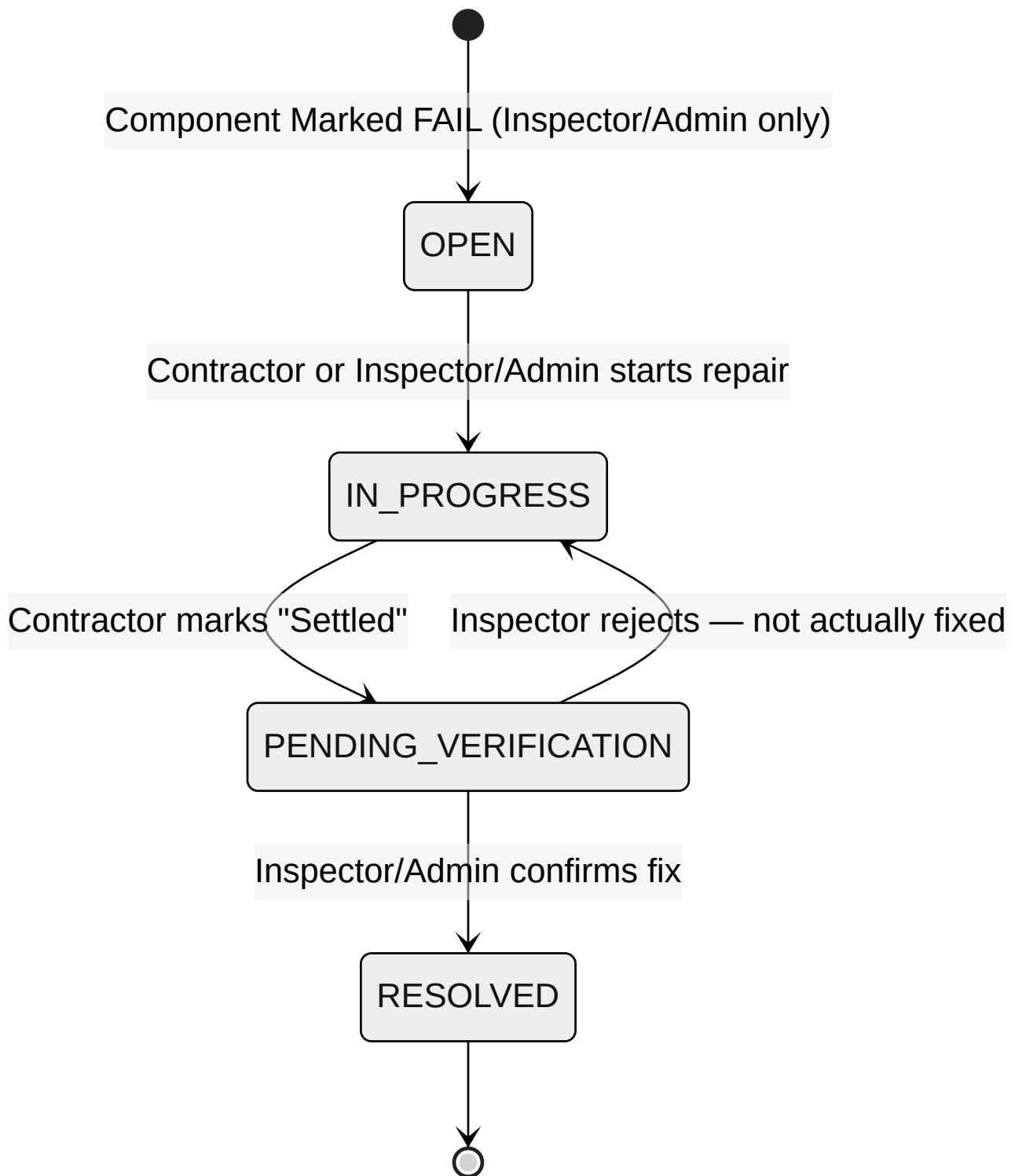
    Admin->>Admin: 1. Create Project, set GFA (incl. assign Inspector & Contractor)
    Note over Admin: System auto-generates N sample slots
    Admin->>Admin: 2. Configure sample room location names & Save — pauses here for Admin
    Note over Admin: Project now appears under "My Assigned Projects"
    Admin->>Inspector: 3. Inspector opens project & runs Components Grid Inspection (Cards View Mode)
    Note over Inspector: FAIL checks auto-spawn OPEN Defects with Spatie Media photos
    Admin->>Contractor: 4. Defect now appears under Contractor's "My Defectors"
    Contractor->>Contractor: 5. Contractor marks "Start Repair" (IN_PROGRESS), then "Mark Settled" (PENDING_VERIFICATION) once fixed
    Contractor->>Inspector: 6. Inspector re-inspects & either Confirms Resolved or Rejects back to IN_PROGRESS
    Note over Inspector: 7. System recalculates Final G-IDS Score & Rating (GOOD/MODERATE/WEAK) — Inspector is notified once inspection is 100% done and all defects are resolved
    Admin->>Admin: 8. Admin clicks "Mark as Completed" (only enabled once inspection is done and every defect is resolved), then exports the Signed G-IDS PDF Certificate
```



The project page's Case Ledger is the single place this whole handoff is tracked — each row shows who's responsible, the key metric, and one action. Here, Field Inspection is active (green) and Defect Rectification needs attention (red) — Final Certificate stays locked until both clear.

4. Defect Lifecycle & Status Workflow

When a defect is discovered during site inspection, it now follows a **4-stage** lifecycle, splitting the old single "In Progress" phase into a self-reported repair step and a formal verification step:



Defect Status Breakdown

! Important

Who updates defect status? The Contractor has their own login and updates status themselves for the repair steps (`IN_PROGRESS` , `PENDING_VERIFICATION`). The final `RESOLVED` confirmation — and the ability to reject a repair back to `IN_PROGRESS` — remains exclusively with the **Inspector** or **Admin**. A Contractor can never self-certify their own work as fully resolved.

1. ● OPEN (Newly Discovered Defect)

- **Trigger:** Automatically created when an Inspector (or Admin) marks any of the 5 component criteria (Finishing, Hollow, Levelling, Joint, Crack) as **FAIL** during inspection, or manually logged.
- **Action:** Defect appears in the assigned Contractor's **My Defects** list and the office's **Defects Register**, with severity tag (**Low** , **Medium** , **High**), room location, component code, and photo evidence via Spatie Media Library.

2. ● IN_PROGRESS (Repair Underway)

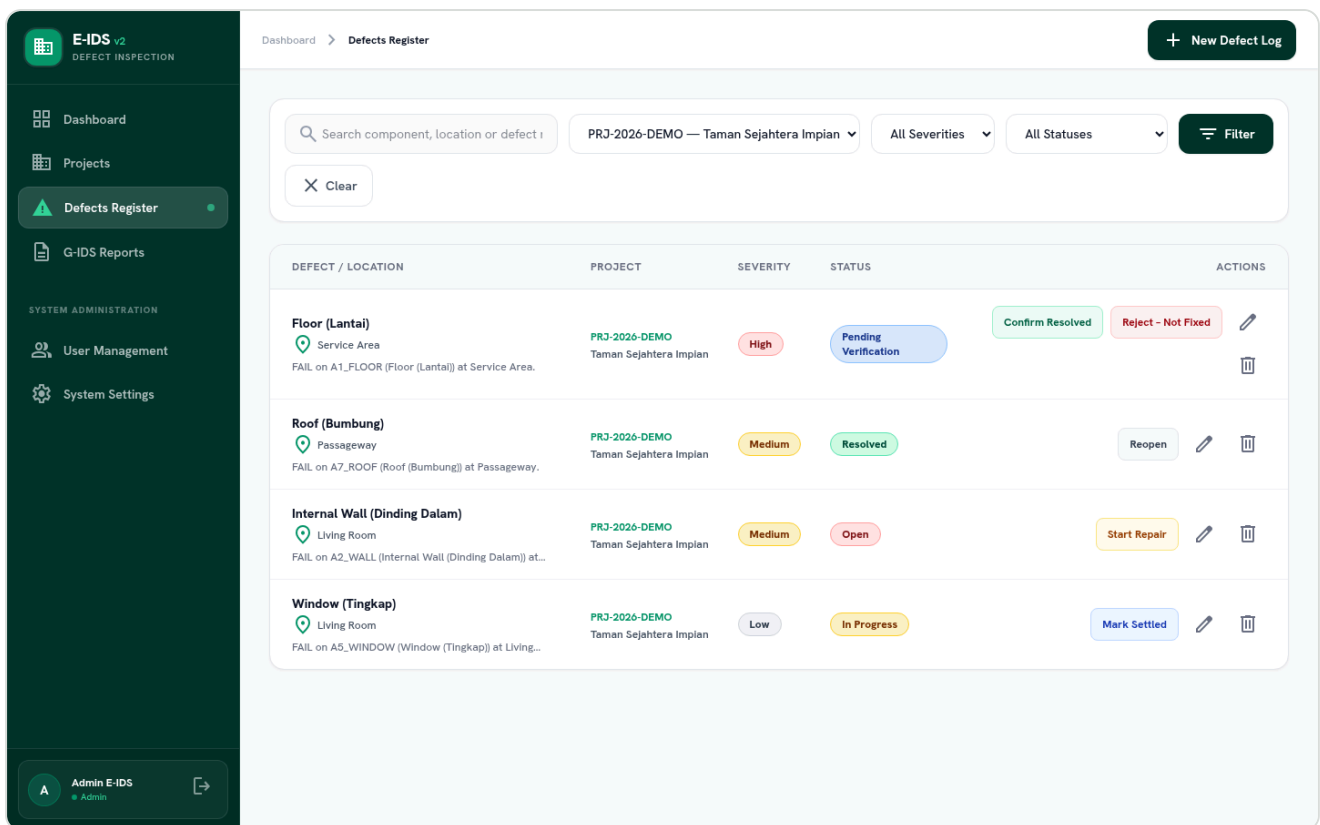
- **Trigger:** The Contractor logs in and clicks **Start Repair** on the defect once they begin work on site (e.g., plaster touch-up, tile replacement, re-levelling). An Inspector or Admin can also set this manually if needed.
- **Action:** Status updates in real time so the office knows a repair is genuinely underway, not just reported informally.

3. ● PENDING_VERIFICATION (Awaiting Inspector Sign-off)

- **Trigger:** The Contractor clicks **Mark Settled** once they believe the repair is complete.
- **Action:** The defect moves into a queue awaiting the Inspector's physical re-check. The Contractor sees a read-only "Awaiting Verification" badge and cannot advance it further themselves.

4. ● RESOLVED (Verified & Closed)

- **Trigger:** The Inspector (or Admin) conducts a re-inspection of the specified location on site.
- **Action:** Upon verifying the defect has been properly rectified to G-IDS standards, they click **Confirm Resolved**. If the repair is inadequate, they instead click **Reject – Not Fixed**, sending it back to **IN_PROGRESS** for the Contractor to redo.



All four states side by side: Open shows Start Repair, In Progress shows Mark Settled, Pending Verification shows Confirm Resolved / Reject – Not Fixed, and Resolved shows Reopen.

5. UI/UX Feature Highlights

Role-Aware Guidance

Every project page shows a one-line **"what's next"** banner tailored to the viewer's role — e.g. an Admin sees *"Waiting on Inspector Aiman to begin the Components Grid inspection"*, while an Inspector sees *"2 defect(s) awaiting your verification."* The Dashboard additionally surfaces an **Action Required** widget for Admin/Inspector, listing their own visible projects that still need attention.

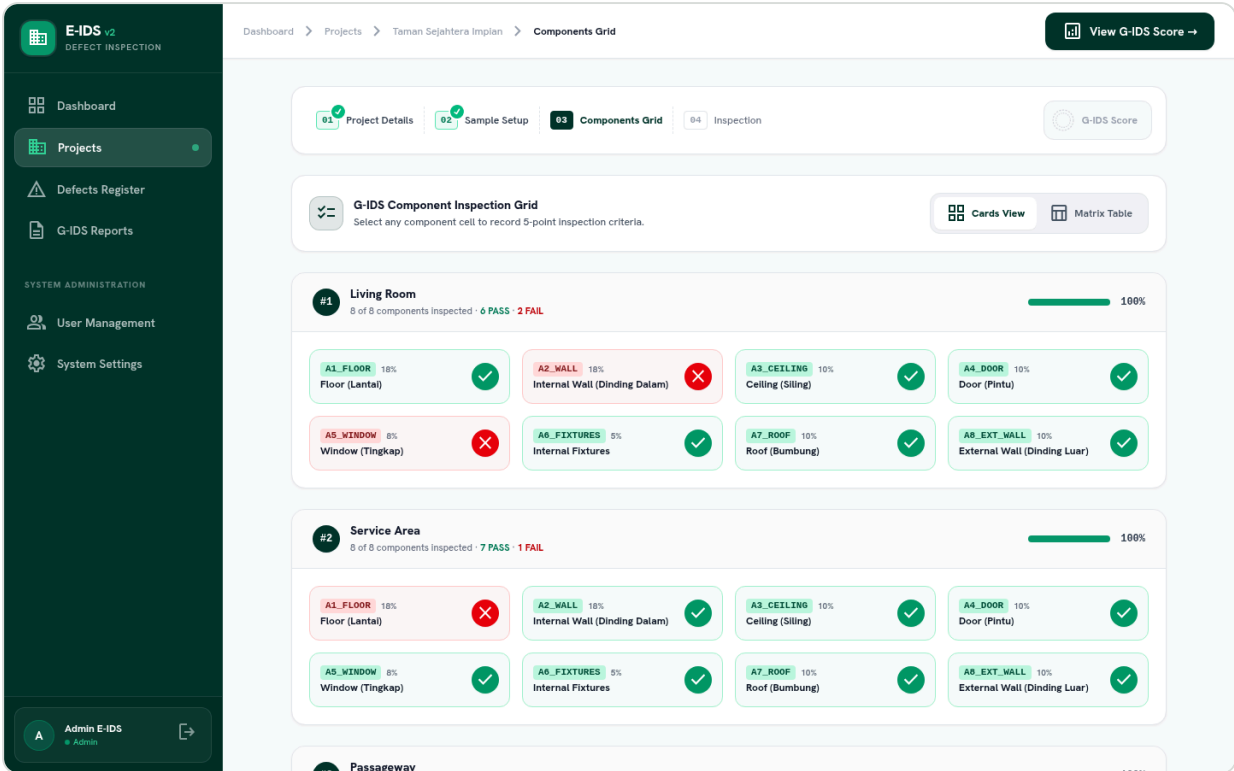
Dual View Mode (Components Grid)

To eliminate horizontal side-scrolling on mobile and tablet screens during site walkthroughs, E-IDS v2 provides two view modes on the **Components Grid**:

1. Cards View (Default — Mobile & Touch Optimized):

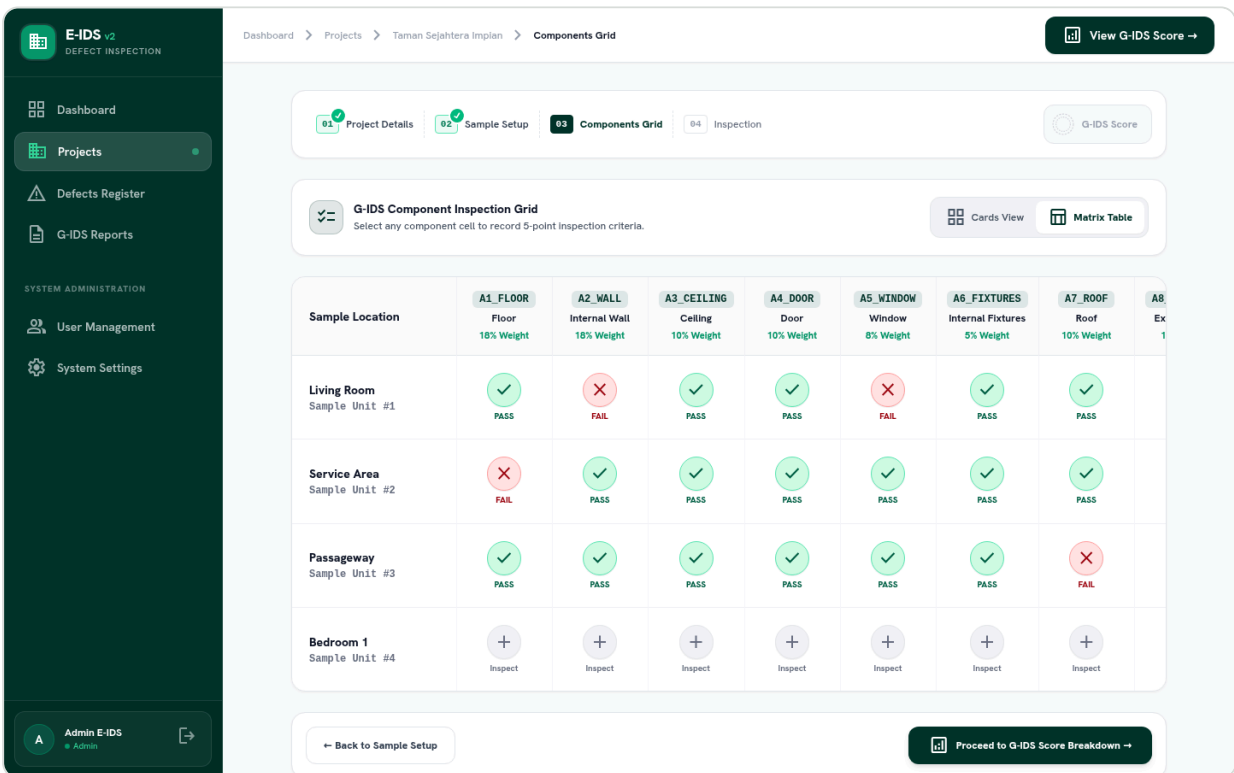
- Sample units are stacked as vertical cards.
- 8 architectural components are rendered in a 2 to 4-column touch grid with high-contrast **PASS**, **FAIL**, or **PENDING** badges.

- **Zero side-scrolling required.**



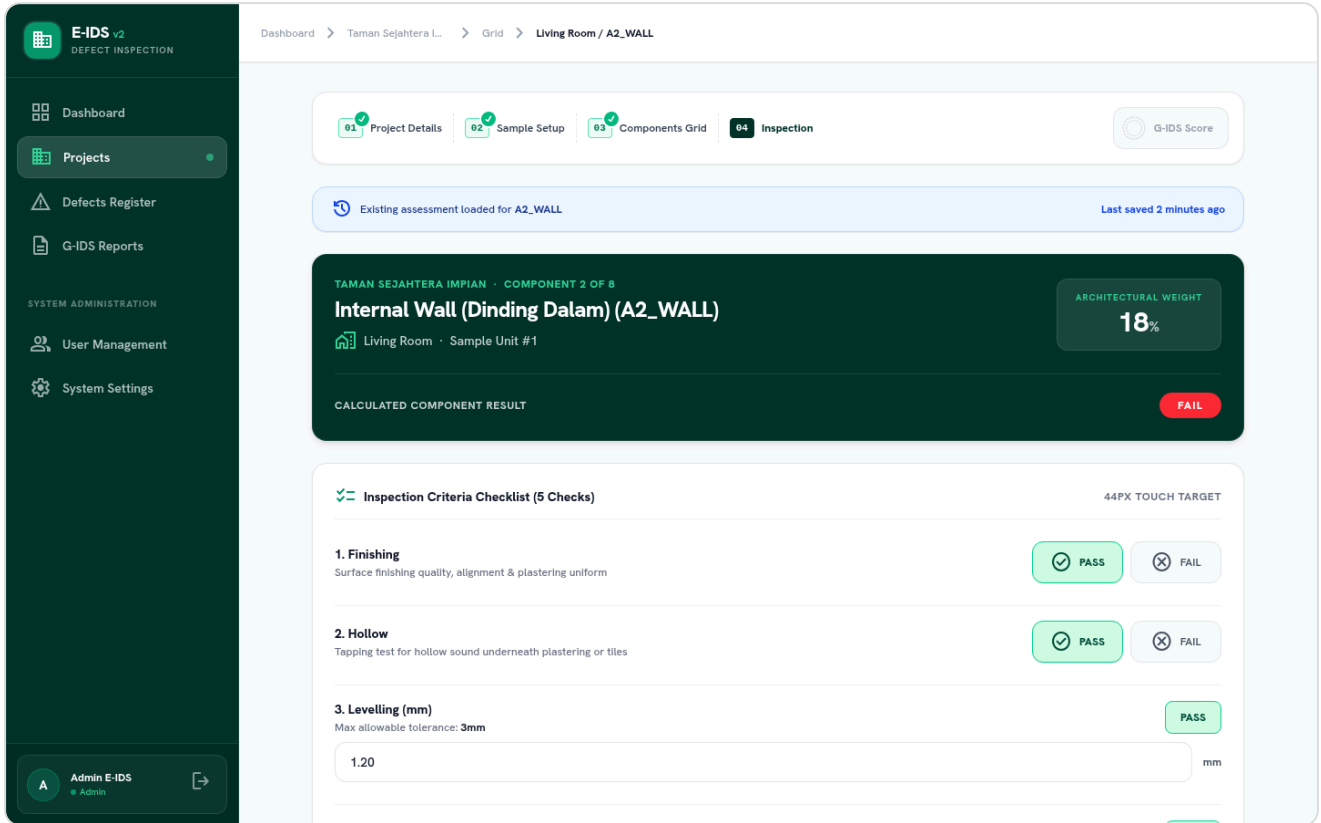
2. Matrix Table View (Spreadsheet Mode):

- Traditional table grid for desktop monitors and office review.



Spatie Media Library Photo Storage

- Automated thumbnail generation (300 × 300px) for fast table rendering.
- HD preview conversion (800 × 600px) for PDF reports.
- Media items attached to inspection assessments are automatically copied to auto-spawned defect records.



The screenshot displays the E-IDS v2 Defect Inspection interface. The left sidebar contains navigation links: Dashboard, Projects, Defects Register, G-IDS Reports, and System Administration (User Management, System Settings). The main content area shows a breadcrumb trail: Dashboard > Taman Sejahtera L... > Grid > Living Room / A2_WALL. A progress bar at the top indicates four steps: 01 Project Details, 02 Sample Setup, 03 Components Grid, and 04 Inspection. A notification states 'Existing assessment loaded for A2_WALL' with a timestamp 'Last saved 2 minutes ago'. The main assessment card for 'Internal Wall (Dinding Dalam) (A2_WALL)' shows an 'ARCHITECTURAL WEIGHT' of 18% and a 'CALCULATED COMPONENT RESULT' of 'FAIL'. Below this is an 'Inspection Criteria Checklist (5 Checks)' with a '44PX TOUCH TARGET' label. The checklist includes: 1. Finishing (Surface finishing quality, alignment & plastering uniform) with 'PASS' and 'FAIL' buttons; 2. Hollow (Tapping test for hollow sound underneath plastering or tiles) with 'PASS' and 'FAIL' buttons; 3. Levelling (mm) (Max allowable tolerance: 3mm) with a 'PASS' button and a text input field containing '1.20'.

The 5-point assessment form for one component — Finishing, Hollow, Levelling, Joint/Gap, and Crack, with the calculated PASS/FAIL result and photo evidence upload.

Modal Confirmation Dialogs

- All destructive actions (Deleting Projects, Deleting Defects, Deleting User Accounts) are protected by an Alpine.js modal dialog (`x-modal-confirm`) requiring explicit confirmation.

6. G-IDS Scoring Framework

The total G-IDS score (S_{total}) is calculated using native CIS 7 rules:

$$S_{\text{total}} = S_{\text{arch}} + S_{\text{ME}} + S_{\text{external}}$$

- **Architectural Subtotal (S_{arch}):** Sum of 8 weighted component pass rates (S_{comp}):

$$S_{\text{comp}} = \left(\frac{\text{Pass Count}}{\text{Total Assessed}} \right) \times \text{Weightage \%}$$

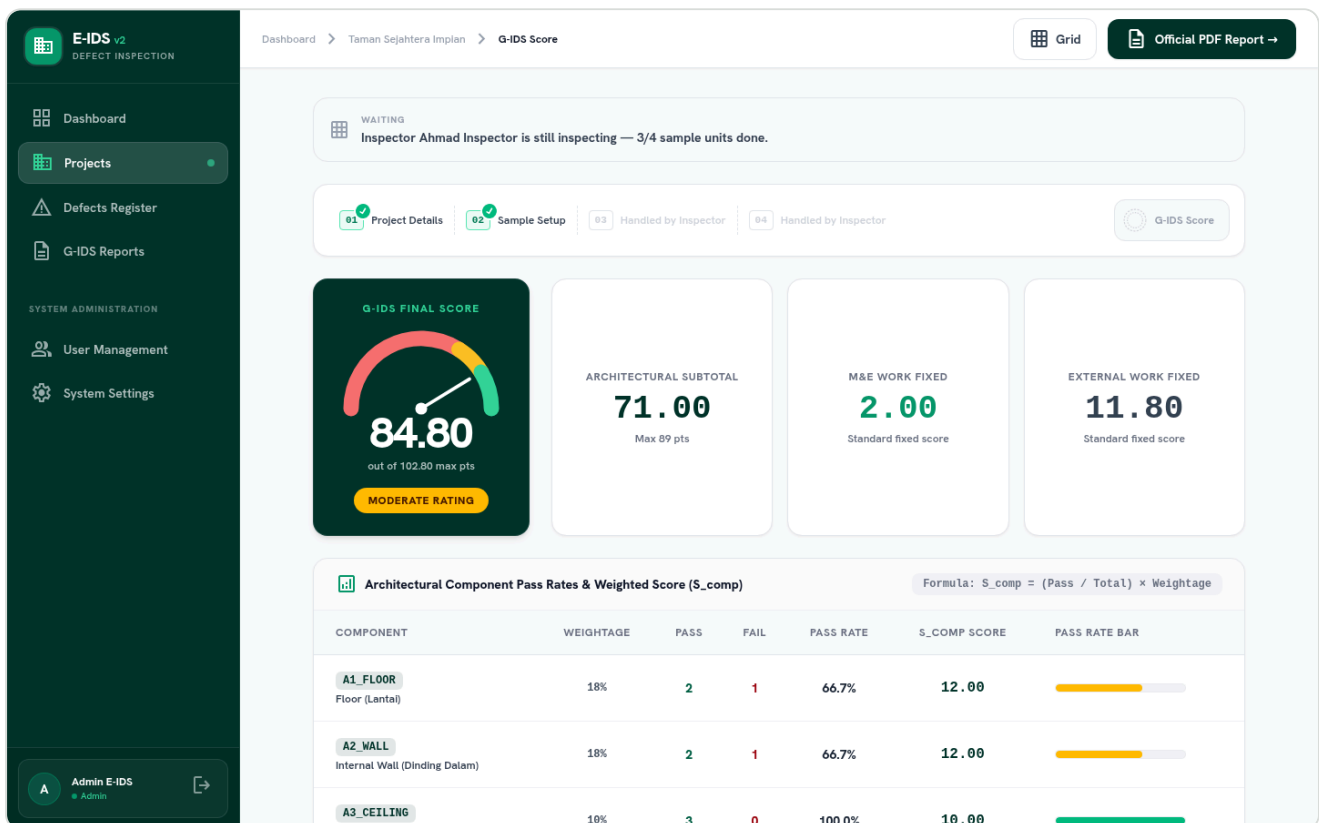
- **M&E Fixed Score (S_{ME}):** 2.00 pts
- **External Work Fixed Score (S_{external}):** 11.80 pts

Rating Thresholds



Note

- **GOOD RATING:** Total score ≥ 85.00 pts
- **MODERATE RATING:** Total score ≥ 70.00 pts and < 85.00 pts
- **WEAK RATING:** Total score < 70.00 pts



The G-IDS Score page: the gauge's needle position and red/amber/green zones map directly to the rating thresholds above, with the full component-by-component breakdown below.

7. Step-by-Step Operating Instructions

Step 1: Project Registration & Team Assignment (Admin)

1. Navigate to **Projects** → Click **+ New Project**.
2. Enter Project Ref No., Project Name, Developer, Contractor, Building Type, and Total Units.
3. Enter Gross Floor Area (GFA m^2). The system auto-calculates sample units ($N = \lceil \text{GFA} \div 60 \rceil$).
4. Select the **Assigned Inspector** and **Assigned Contractor** from the dropdowns. (An Inspector creating a project instead is auto-assigned to themselves, and can still choose the Contractor — only Admin can assign the Inspector field.)
5. Click **Create Project**.

E-IDS v2

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Projects > New Project

Project Specifications & Metadata

PROJECT REFERENCE NO. *

e.g. PRJ-2026-001

PROJECT NAME *

e.g. Taman Merlimau Perdana

SITE LOCATION / ADDRESS

e.g. Merlimau, Melaka

DEVELOPER NAME *

Developer company name

CONTRACTOR NAME *

Contractor company name

BUILDING TYPE *

Select building type...

INITIAL STATUS *

In Inspection

ASSIGNED INSPECTOR

Unassigned (Open for all inspectors)

ASSIGNED CONTRACTOR

Unassigned

TOTAL PROJECT UNITS *

1

Sample Unit Calculation

Formula: $N = \max(1, \text{cell}(\text{GFA} + 60))$

GROSS FLOOR AREA (GFA M²) *

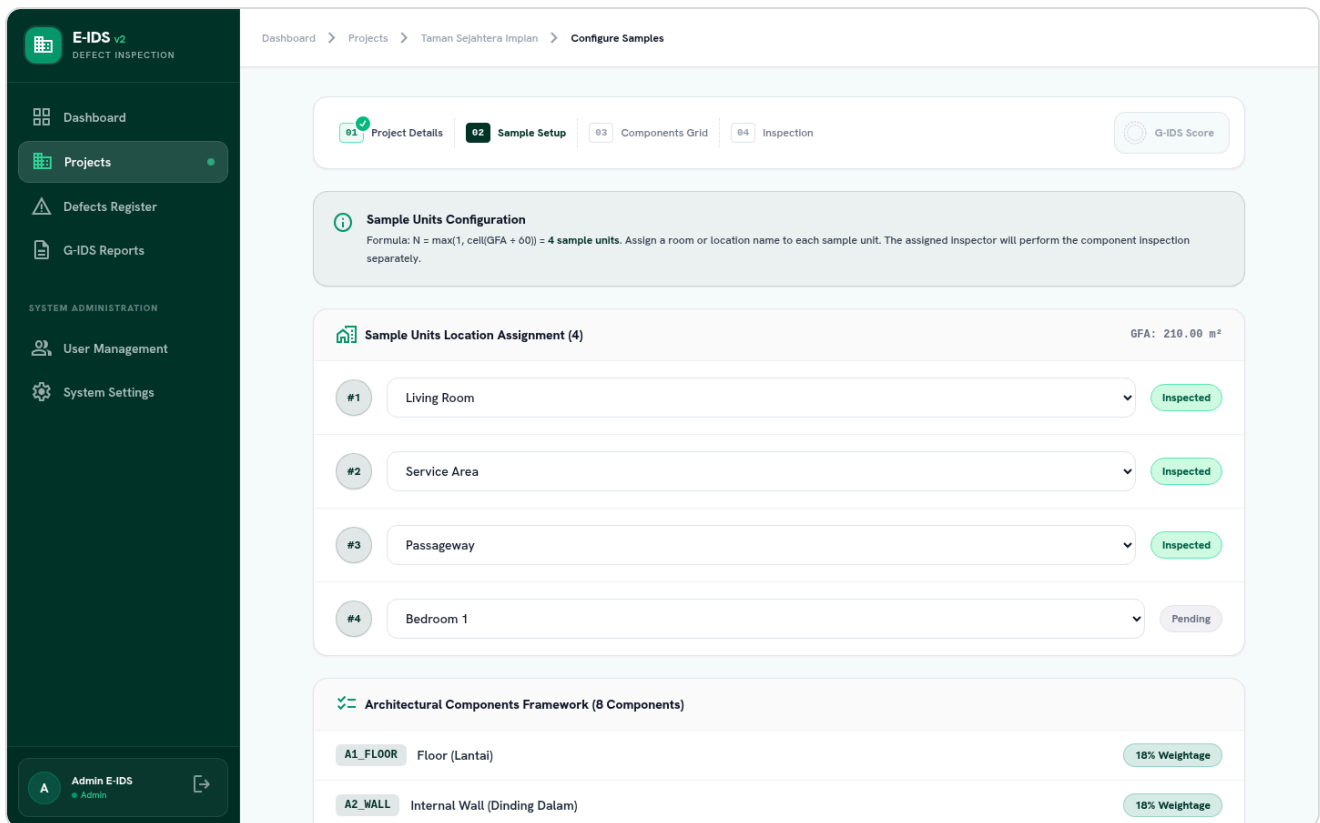
e.g. 210.00

CALCULATED SAMPLE UNITS (N)

0

Step 2: Sample Location Setup (Admin / Inspector)

1. System automatically redirects to **Sample Setup**.
2. Select default location names (e.g. *Master Bedroom, Living Room, Kitchen*) or enter custom room names.
3. Click **Save Locations**. For Admin, this pauses their part of setup — you're returned to the Project Overview page. The project now appears under the assigned Inspector's **My Assigned Projects**, ready for on-site inspection. (An Inspector doing their own setup instead continues straight into the Components Grid.)



Step 3: Component Inspection Matrix Grid (Inspector)

1. Open **Components Grid** (defaults to mobile-friendly **Cards View**) from **My Assigned Projects**.
2. Click any component tile (**D1** , **D2** , etc.) to open the **5-Point Assessment Form**:
 - **Finishing**: PASS / FAIL
 - **Hollow**: PASS / FAIL
 - **Levelling (mm)**: Enter measured mm (auto-checked against $\leq 3.0\text{mm}$ tolerance)
 - **Joint/Gap (mm)**: Enter measured mm (auto-checked against $\leq 1.0\text{mm}$ tolerance)
 - **Crack**: PASS / FAIL
3. Upload photo evidence if defects exist.
4. Click **Save & Next** to advance through component slots smoothly. A **FAIL** automatically opens a defect in the Contractor's queue.

Step 4: Repair & Verification (Contractor, then Inspector)

1. **Contractor** logs in and lands directly on **My Defects** — a list scoped to their one assigned project.
2. Click **Start Repair** on an **OPEN** defect once work begins on site (moves it to **IN_PROGRESS**).
3. Click **Mark Settled** once the repair is complete (moves it to **PENDING_VERIFICATION**).
4. **Inspector** opens the **Defects Register**, filters to **Pending Verification** , and re-inspects the location on site.
5. Inspector clicks **Confirm Resolved** if the fix holds up, or **Reject – Not Fixed** to send it back to **IN_PROGRESS** for another attempt.

Step 5: Score Review (Admin / Inspector)

1. Once every sample unit is inspected, the certification-seal button in the step bar unlocks — click it, or **G-IDS Score**, to review the final score, rating, and architectural component pass rates.
2. If any defect is still **OPEN**, **IN_PROGRESS**, or **PENDING_VERIFICATION**, the score page and PDF report stay locked with an explanatory message until every defect reaches **RESOLVED**.

Step 6: Completion Sign-off & PDF Export (Admin only)

1. Once inspection is 100% done and every defect is **RESOLVED**, the project page's **Final Certificate** panel shows a **Mark as Completed** button.
2. Click it to formally close out the project (status becomes **Completed**) — this is a deliberate action; nothing marks a project Completed automatically.
3. Click **Generate Official PDF Report** to view or print the formal signed G-IDS Inspection Certificate.

FORMAL G-IDS INSPECTION CERTIFICATE

Desa Aman Villa
PRJ-2026-CERT · Semi-D
Alor Gajah, Melaka

FINAL G-IDS SCORE
102.80
GOOD RATING

89.00 Architectural Subtotal
2.00 M&E Work (Fixed)
11.80 External Work (Fixed)
0 Total Defects Logged

Project Information & Audit Metadata

DEVELOPER Impian Setia Sdn Bhd	CONTRACTOR Kukuh Sentosa Construction Sdn Bhd	TOTAL UNITS 20 units
GROSS FLOOR AREA (GFA) 130.00 m²	LEAD INSPECTOR Admin E-IDS	AUDIT COMPLETION DATE 02 Aug 2026

Architectural Component Score Breakdown
S_comp = (Pass / Total) × Weightage

COMPONENT	WEIGHTAGE	PASS	FAIL	PASS RATE	S_COMP SCORE
A1_FLOOR Floor (Lantai)	18%	2	0	100.0%	18.00
A2_WALL Internal Wall (Dinding Dalam)	18%	2	0	100.0%	18.00
A3_CEILING Ceiling (Silang)	10%	2	0	100.0%	10.00

The signed certificate — only reachable once inspection is 100% done and every defect is **RESOLVED**. Its "Export Signed PDF Certificate" button in the top bar produces the downloadable file.

8. Demo / Testing Accounts

Running `docker exec -w /var/www eidsv2 php artisan migrate:fresh --seed` loads a ready-to-test dataset: **2 projects and 5 accounts covering all 3 roles**, split into an "A" set (Project 1) and a "B"

set (Project 2) so you can log in as different users and see the role-scoping working directly — e.g. Inspector A only ever sees Project 1, Inspector B only ever sees Project 2.

ROLE	ACCOUNT	PASSWORD	SCOPE
Admin	admin@eids.gov.my	password	Sees everything; created both projects
Inspector	inspector.a@eids.gov.my	password	Assigned to Project 1 only
Inspector	inspector.b@eids.gov.my	password	Assigned to Project 2 only
Contractor	contractor.a@eids.gov.my	password	Assigned to Project 1 only — lands on My Defects
Contractor	contractor.b@eids.gov.my	password	Assigned to Project 2 only — lands on My Defects

Project 1 — "Taman Merlimau Perdana" (PRJ-2026-001): 4 sample rooms, 3 fully inspected + 1 left pending (so you can see the "Continue — 3/4 done" banner). One defect seeded in **every lifecycle state** so every action button can be tested immediately:

DEFECT	LOCATION / COMPONENT	STATUS	WHAT TO TEST
1	Living Room / Internal Wall	OPEN	Contractor A sees Start Repair
2	Living Room / Window	IN_PROGRESS	Contractor A sees Mark Settled
3	Service Area / Floor	PENDING_VERIFICATION	Inspector A sees Confirm Resolved / Reject ; Contractor A sees a read-only "Awaiting Verification" badge
4	Passageway / Roof	RESOLVED	Inspector A sees Reopen

Score: 84.80 — **MODERATE** (just under the GOOD threshold, useful for demonstrating the rating boundary).

Project 2 — "Desa Aman Villa" (PRJ-2026-002): smaller, minimal setup — one **OPEN** defect only. Its main purpose is proving Contractor B / Inspector B never see Project 1's data. Score: 92.80 — **GOOD**.